

Title:

AI-Powered Multi-Agent Smart Learning Platform Using LangChain, LLMs, and Video-Based Content Analysis

Abstract:

This project presents an AI-powered Smart Learning Platform that enhances the peer-to-peer learning experience by integrating multi-agent Large Language Model (LLM) intelligence with a video-based educational website. The system analyses lesson video titles and description of the video, generates expected semester examination questions, creates topic-wise quizzes, and provides interactive chatbot support to students. A Python-based multi-agent architecture is developed using LangChain and Gemini/OpenAI LLMs, incorporating three core algorithms: the ReAct (Reason + Act) algorithm for step-by-step reasoning and decision-making, the CAMEL Multi-Agent Collaboration algorithm for structured interaction among specialized agents, and Retrieval-Augmented Generation (RAG) with vector similarity search for accurate, context-aware retrieval of syllabus content, circulars, and lecture transcripts.

Each agent performs specific tasks such as video content analysis, question generation, quiz creation, document retrieval, and conversational assistance, working together through coordinated reasoning powered by the ReAct and CAMEL frameworks. To integrate the AI with the existing Java Spring Boot web application, a dedicated Python microservice is built using FastAPI, which exposes REST APIs for communication between the website and the AI engine. FastAPI manages request handling, model execution, and agent orchestration, enabling seamless interaction between the frontend and the AI modules. The platform also employs a vector database to support the RAG pipeline for high-accuracy responses grounded in academic materials.

The final system provides personalized learning support by automatically generating academic content, offering intelligent Q&A assistance, and enhancing self-paced learning. This hybrid architecture—combining Spring Boot, frontend, FastAPI microservices, and multi-agent LLM pipelines—creates a modern, scalable, and AI-driven educational platform suitable for real-world academic environments.